The brief

Microsoft AssessFlow

An access-governance application where **AI advises, people authorise,** and every request remains accountable.

FULL-STACK ENGINEERING POLICY-GROUNDED AI ACCESS GOVERNANCE

Executive perspective

I engineered AssessFlow as a working internal application for employee access requests. A React operational console and FastAPI services coordinate persistent PostgreSQL records, policy evidence, structured advisory assessments, assigned human approvals, and manual fulfilment. The **backend - not the model - owns permissions, state transitions, and decisions.** Microsoft integrations are implemented as source-controlled adapters and contracts; real tenant activation and cloud deployment remain separate, authorised steps.

The operational journey

Request and clarify

Employees submit their own application-access requests, supply missing information on the same request ID, track status and cancel where allowed.

Evidence before recommendation

Current policy versions, immutable citations, departmental access controls and deterministic eligibility rules shape the advisory output.

Approve with authority

Assigned approvers or administrators give reasoned decisions. Self-approval and duplicate decisions are blocked by application rules.

Fulfil with an audit trail

Approved requests create manual service tasks. Operators record verified fulfilment references; the application does not change directory permissions.

The assessment advises. People decide.

The main engineering decision is a hard separation between policy interpretation, administrative authority and recorded fulfilment. A model recommendation cannot grant access.

The request path

Employee

Submit or clarify an access request.

Policy and AI

Redact, retrieve, assess, cite.

Approver

Review and decide with a reason.

Operator

Record the manual fulfilment.

Four governing boundaries

Versioned, permission-aware evidence

UTF-8 policy text is versioned, digested and chunked. Retrieval respects department access and rechecks source-of-truth versions before citing evidence.

Advisory output is subordinate

Privacy redaction, strict schemas, citation validation and deterministic overrides protect the workflow when assessment evidence is incomplete, conflicting or expired.

Restricted employee tools

Copilot Studio source instructions and OAuth connector contracts expose create, status, clarification and permitted-policy search. They do not grant or fulfil access.

Signed workflow events

Power Automate integration contracts validate callback signatures, timestamp, replay and the assigned actor. No external approval result bypasses the API.

Technical spine: React and Vite / FastAPI and Pydantic / SQLAlchemy and Alembic / PostgreSQL / Microsoft adapter boundaries

Verified locally.
Azure activation pending.

The recorded tests establish substantial local application and release-engineering work. They do not establish live Microsoft provider execution, published Copilot functionality, or production availability.

65
PostgreSQL-backed backend tests

33
Release-safety tests

6
Frontend component tests

3
Playwright browser workflows

13 / 13
Offline assessment and workflow scenarios passed. This was a deterministic simulation; **no live AI model was invoked.**

Implemented and exercised

Persisted React/FastAPI request workflows and roles; PostgreSQL transactions, migrations and concurrency checks; policy citations; deterministic simulation; Microsoft service adapters and contracts; Bicep compilation; container builds; four successful hosted GitHub Actions CI jobs.

Prepared, not activated

No live tenant sign-in, Copilot Studio publication, Foundry/Azure OpenAI inference, Language PII, AI Search, Graph request, Power Automate flow or Azure staging promotion. Deployment automation is prepared, not run. No paid resources were provisioned.

Evidence and source

Source: github.com/keenosmith-del/ms-assessflow | Main reviewed: **1ba858f** | Hosted CI evidence: **e1bea3e**

Engineering: docs/portfolio/engineering.md | Verification: docs/testing/results.md | Release: docs/iterations/iteration-008.md

Release status: local system verified; Azure staging configuration, independent approvals and live integration remain outstanding.